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Chapter 8

Index Models

INVESTMENTS
THIRTEENTH EDITION
BODIE, KANE, MARCUS

Chapter Overview

- Drawbacks to Markowitz procedure.
 - Covariance matrix requires a huge number of estimates.
 - Model does not provide any guidelines for finding useful estimates of these covariances or the risk premiums.
- Introduction of index models.
 - Simplifies estimation of the covariance matrix.
 - Enhances analysis of security risk premiums.
- Optimal risky portfolios constructed using the index model.

A Single-Factor Security Market ¹

- Advantages of the single-index model.
 - Number of estimates required is a small fraction of what would otherwise be needed.
 - Specialization of effort in security analysis.

A Single-Factor Security Market ²

- Systematic versus Firm-Specific Risk
 - We can decompose the rate of return on any security, i , into the sum of its expected plus unanticipated components:

$$r_i = E(r_i) + e_i$$

- Then we can decompose the sources of uncertainty into uncertainty about the economy as a whole, which is captured by m , and uncertainty about the firm in particular, which is captured by e_i

$$r_i = E(r_i) + m + e_i$$

A Single-Factor Security Market ³

- Finally, we recognized that some securities will be more sensitive than others to macro-economic shock

$$r_i = E(r_i) + \beta_i m + e_i$$

- β_i = sensitivity coefficient for firm i
- m = market factor that measures unanticipated developments in the macroeconomy
- e_i = firm-specific random variable

Single-Index Model ¹

- Regression equation (time series)

(notation Excess return format: $R_i = r_i - r_f$):

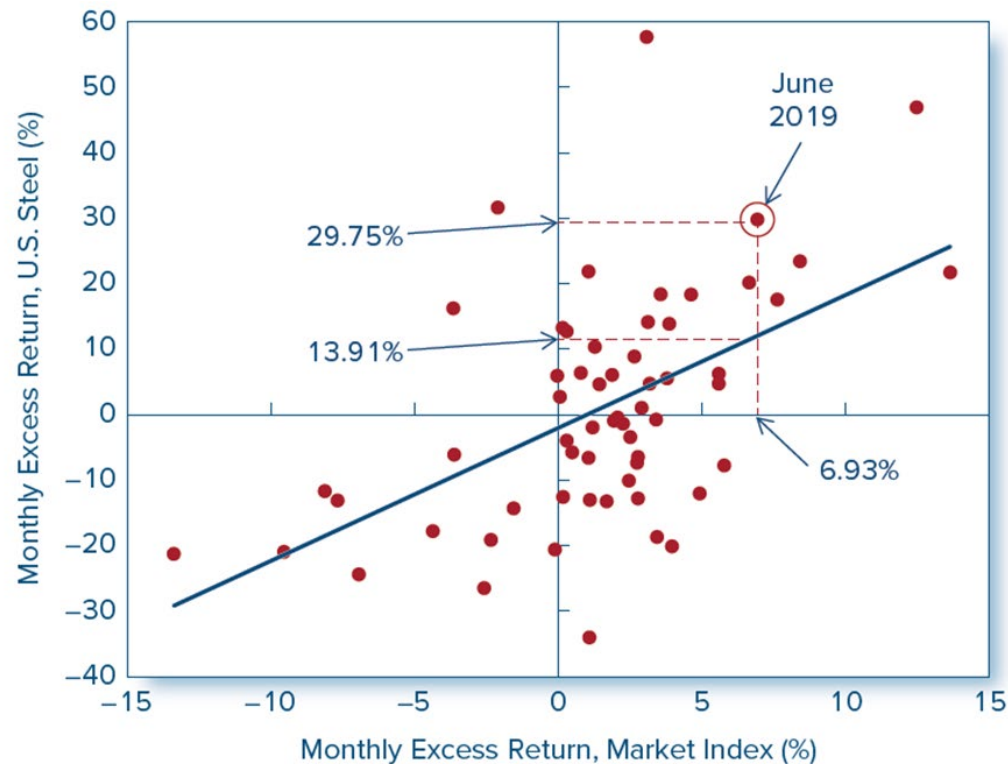
$$R_i(t) = \alpha_i + \beta_i R_M(t) + e_i(t)$$

- Expected return–beta relationship.

$$E(R_i) = \alpha_i + \beta_i E(R_M)$$

α &
CAPM

Figure 8.1 Scatter Diagram



$$R_{U.S.Steel}(t) = \alpha_{U.S.Steel} + \beta_{U.S.Steel}R_{index}(t) + e_{U.S.Steel}(t)$$

Figure 8.1 Scatter diagram for U.S. Steel against the market index and U.S. Steel's security characteristic line, 5 years ending December 2021.

Single-Index Model ₂

Total risk = Systematic risk + Firm-specific risk.

$$\sigma_i^2 = \underbrace{\beta_i^2 \sigma_M^2}_{\text{Systematic risk}} + \underbrace{\sigma^2(e_i)}_{\text{Firm-specific risk}}$$

Covariance = Product of betas $\times$ Market-index risk.

$$\text{Cov}(r_i, r_j) = \beta_i \beta_j \sigma_M^2$$

Correlation = Product of correlations with the market index.

$$\begin{aligned} \text{Corr}(r_i, r_j) &= \frac{\beta_i \beta_j \sigma_M^2}{\sigma_i \sigma_j} = \frac{\beta_i \sigma_M^2 \beta_j \sigma_M^2}{\sigma_i \sigma_M \sigma_j \sigma_M} \\ &= \underbrace{\text{Corr}(r_i, r_M)}_{\text{Correlation of } r_i \text{ with market}} \times \underbrace{\text{Corr}(r_j, r_M)}_{\text{Correlation of } r_j \text{ with market}} \end{aligned}$$

The Set of Estimates Needed for the Single-Index Model

	Symbol
1. The stock's expected return if the market is neutral, that is, if the market's excess return, $r_M - r_f$ is zero	α_i
2. The component of return due to movements in the overall market in any period; β_i is the security's responsiveness to market movements	$\beta_i(r_M - r_f)$
3. The unexpected component of return in any period due to unexpected events that are relevant only to this security (firm specific)	e_i
4. The variance attributable to the uncertainty of the common macro-economic factor	$\beta_i^2 \sigma_M^2$
5. The variance attributable to firm-specific uncertainty	$\sigma^2(e_i)$

Input List and the Sets of Estimates Needed

Symbol	# of estimates
α_i	n
β_i	n
$\sigma^2(\mathbf{e}_i)$	n
$(r_M - r_f)$	1
σ_M^2	1

$(3n+2)$ versus $n+n+(n^2-n)/2$ estimates needed in the Markowitz model

e.g. If $n=50$, 152 v.s. 1,325 estimates

Index Model and Diversification ¹

$$R_P = \sum_{i=1}^n w_i R_i = \frac{1}{n} \sum_{i=1}^n (\alpha_i + \beta_i R_M + e_i)$$
$$= \alpha_P + \beta_P R_M + e_P$$

**Assume
equally weighted portfolio**

$$\beta_P = \frac{1}{n} \sum_{i=1}^n \beta_i$$

$$\alpha_P = \frac{1}{n} \sum_{i=1}^n \alpha_i$$

$$e_P = \frac{1}{n} \sum_{i=1}^n e_i$$

$$\sigma_P^2 = \beta_P^2 \sigma_M^2 + \sigma^2(e_P)$$

Index Model and Diversification ²

- Portfolio Variance (equal weight) is:

$$\sigma_P^2 = \beta_P^2 \sigma_M^2 + \sigma^2(e_p)$$

$\beta_P^2 \sigma_M^2 =$ Systematic risk of the portfolio
(depends on average beta of the individual securities)

This part of the risk depends on portfolio beta and σ_M^2 and will persist regardless of the extent of portfolio diversification.

$\sigma^2(e_p) =$ Firm-specific risk of the portfolio

Index Model and Diversification ³

- Variance of the equally weighted portfolio of firm-specific components:

$$\sigma^2(e_p) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n e_i\right) = \sum_{i=1}^n \left(\frac{1}{n}\right)^2 \sigma^2(e_i) = \frac{1}{n} \sum_{i=1}^n \frac{\sigma^2(e_i)}{n} = \frac{1}{n} \bar{\sigma}^2(e)$$

- When n gets large, $\sigma^2(e_p)$ becomes negligible.
- As diversification increases, the total variance of a portfolio approaches the systematic variance.

Figure 8.3 The Variance of an Equally Weighted Portfolio With Risk Coefficient, β_p

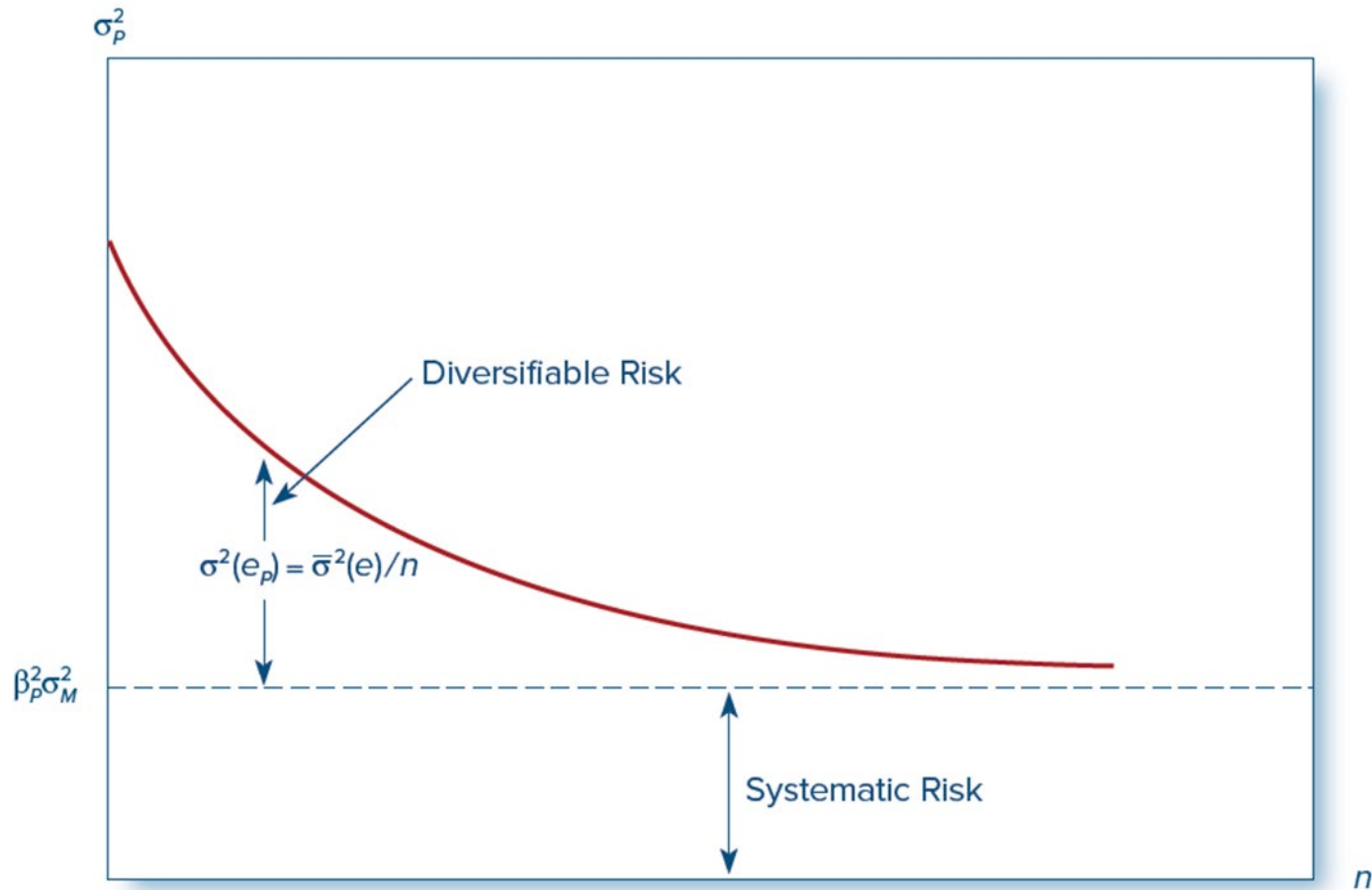


Figure 8.3 Excess Monthly Returns on U.S. Steel and the Market Index

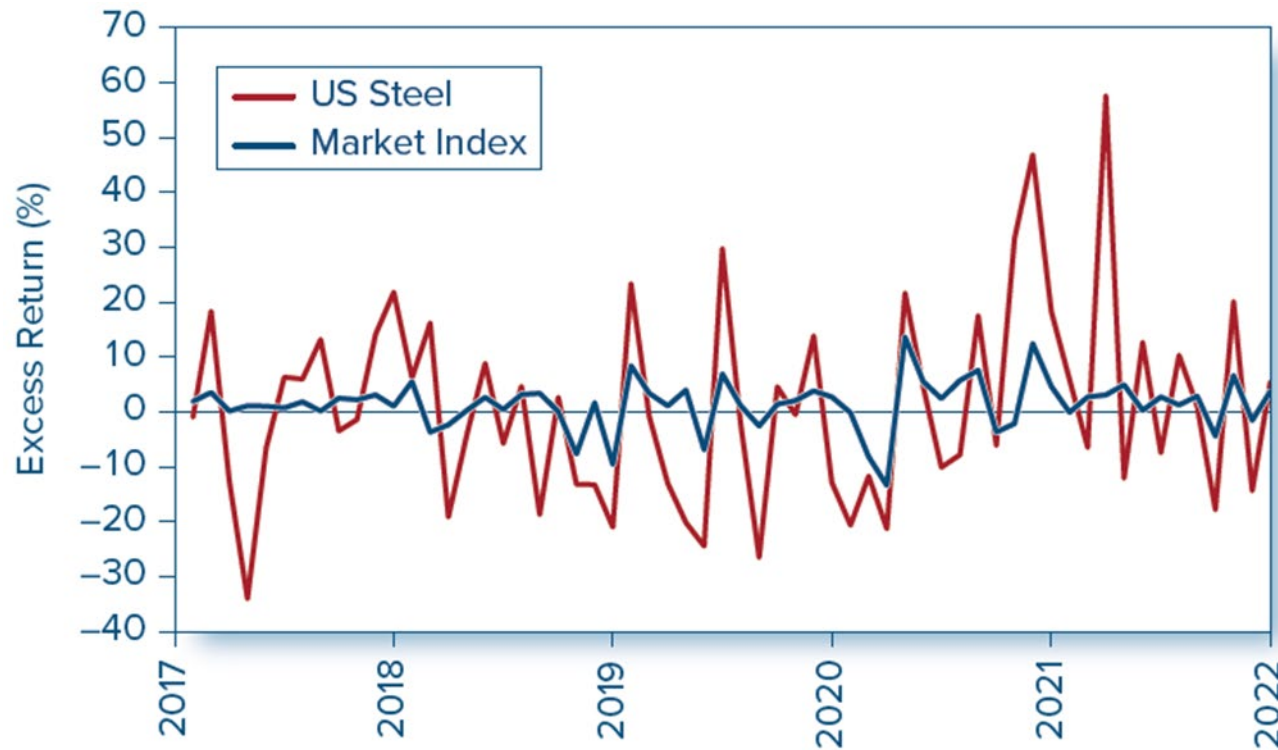


Figure 8.3 Excess monthly returns on U.S. Steel and the market index, 5 years ending December 2021.

Security Characteristic Line (SCL)

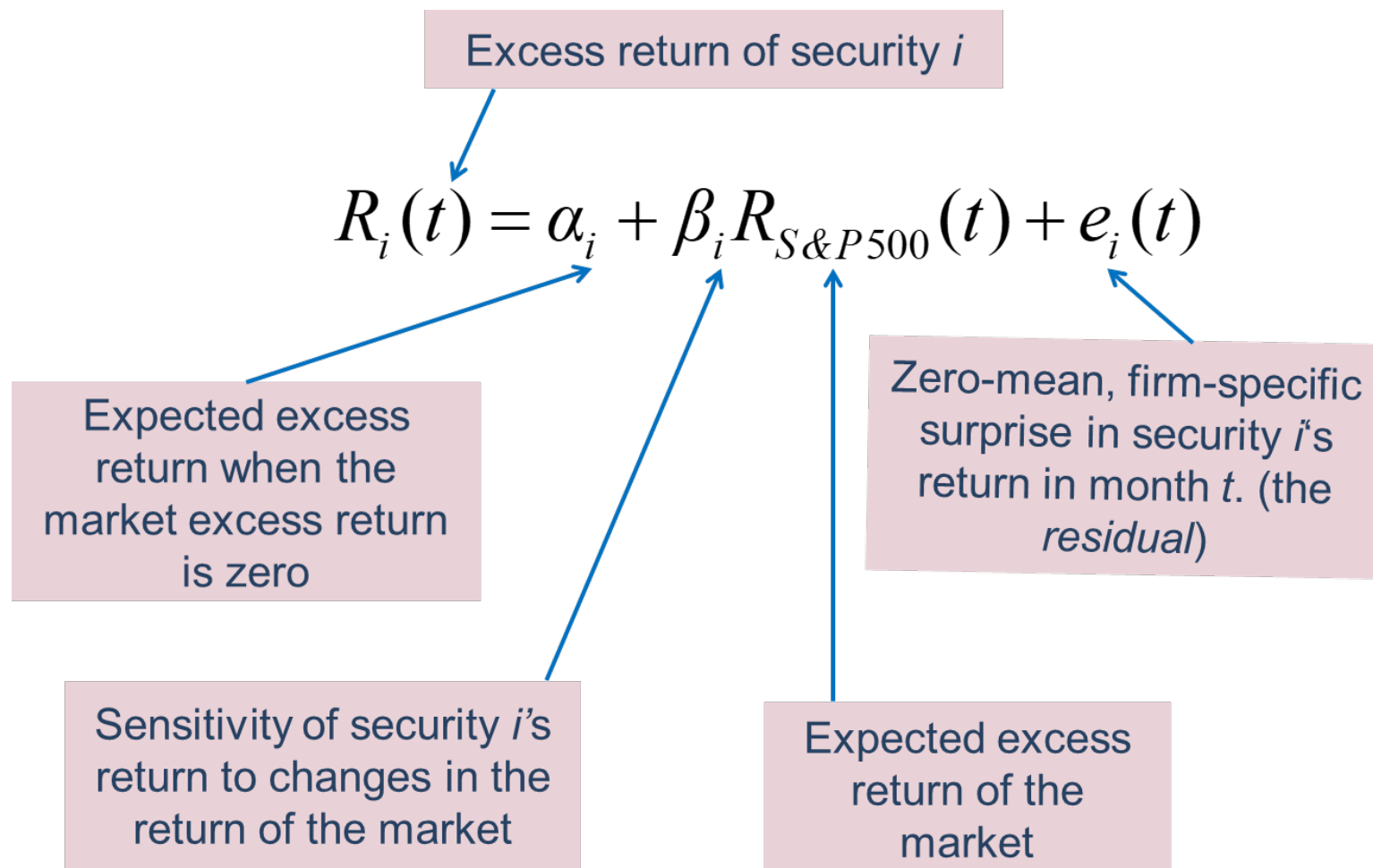


Table 8.1 Excel Output: Regression Statistics

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Table 8.1

Excel output: Regression statistics for U.S. Steel's SCL. Observations are monthly returns measured as percents (not decimals) from January 2017 through December 2021.

Regression Statistics				
Multiple R	0.532			
R-Square	0.283			
Adjusted R-Square	0.271			
Standard Error	15.171			
Observations	60			
Explanatory variable	Coefficient	Standard Error	t-statistic	p-value
Intercept	-2.034	2.052	-0.991	0.326
Market index	2.030	0.424	4.785	0.000

$$R - square = \frac{SSR}{SST} = \frac{\beta_{U.S.Steel}^2 \sigma_{S\&P500}^2}{\underbrace{\beta_{U.S.Steel}^2 \sigma_{S\&P500}^2} + \sigma^2(e_{U.S.Steel})} = 0.283$$

$$Adj R^2 = 1 - (1 - R^2) \frac{n - 1}{n - k - 1}$$

Interpreting the Output

- Correlation of U.S. Steel with the S&P 500 is 0.532
- The model explains about 28.3% of the variation in U.S. Steel's returns
- U.S. Steel's alpha is -2.037% per month, not statistically significant at 5% (p-value=0.326)
- U.S. Steel's beta is 2.030, but the 95% confidence interval is 1.182 to 2.878, which is quite wide

The Industry Version of the Index Model₁

- Use 60 most recent months of price data
- Use S&P 500 as proxy for M
- Compute total returns that ignore dividends
- Estimate index model without excess returns:

$$r = a + br_M + e^*$$

The Industry Version of the Index Model₂

- Practitioners (e.g. Merrill Lynch) use total return, rather than excess returns, in the regression.

$$r = a + br_M + e^*$$

- One can rewrite the equation below to see

$$r - r_f = \alpha + \beta(r_M - r_f) + e$$

$$\begin{aligned} r &= r_f + \alpha + \beta r_M - \beta r_f + e \\ &= [\alpha + r_f(1 - \beta)] + \beta r_M + e \end{aligned}$$

The Industry Version of the Index Model₃

- Several proprietary sources (Merrill Lynch) for such regression results, sometimes called “beta books”.
- Table 8.2
- Note that these alphas are *ex post* (after the fact) measures.
- Security analysis is to forecast alpha values *ex ante* (before the fact)

Table 8.2 Index Model Regressions ¹

Company	Beta	Alpha	R-Square	Residual Std Dev	Standard Error Beta	Standard Error Alpha	Adjusted Beta
Newmont Mining	0.217	0.011	0.016	0.081	0.227	0.011	0.478
Campbell Soup	0.461	-0.008	0.109	0.062	0.173	0.008	0.640
Intel	0.510	0.002	0.107	0.069	0.193	0.009	0.673
McDonald's	0.528	0.008	0.246	0.043	0.121	0.006	0.685
Pfizer	0.615	0.007	0.176	0.062	0.174	0.008	0.743
Coca-Cola	0.644	0.000	0.319	0.044	0.124	0.006	0.762
Starbucks	0.759	0.004	0.291	0.056	0.156	0.008	0.839
Microsoft	0.806	0.019	0.517	0.037	0.102	0.005	0.871
Alphabet (Google)	1.003	0.009	0.523	0.045	0.126	0.006	1.002
Amazon	1.053	0.012	0.372	0.064	0.180	0.009	1.035
Union Pacific	1.126	0.002	0.587	0.044	0.124	0.006	1.084
Walt Disney	1.175	-0.007	0.459	0.060	0.168	0.008	1.117
ExxonMobil	1.295	-0.018	0.491	0.062	0.173	0.008	1.197
Bonk of America	1.442	-0.005	0.662	0.048	0.135	0.007	1.295

Table 8.2 Index Model Regressions ²

Company	Beta	Alpha	R-Square	Residual Std Dev	Standard Error Beta	Standard Error Alpha	Adjusted Beta
Boeing	1.474	-0.009	0.306	0.104	0.291	0.014	1.316
U.S. Steel	2.030	-0.020	0.283	0.152	0.424	0.021	1.687
Marathon Oil	2.986	-0.024	0.477	0.147	0.410	0.020	2.324
AVERAGE	1.066	-0.001	0.350	0.069	0.194	0.009	1.044
STD DEVIATION	0.672	0.012	0.184	0.034	0.095	0.005	0.448

Table 8.2

Index model estimate: Regressions of excess stock returns on the excess return of the broad market index over 60 months, 5 years ending December 2021.

The Industry Version of the Index Model: Predicting Betas ¹

Predicting betas.

- The average beta over all securities is 1; thus, the best forecast of the beta would be that it is 1
- Firms may become more “typical” as they age, causing their betas tend to drift to 1 over time

$$\text{Adjusted beta} = \frac{2}{3} \times \text{Estimated beta} + \frac{1}{3} \times 1.00$$


The Industry Version of the Index Model: Predicting Betas ₂

Forecasting betas.

$$\text{Current beta} = a + b \times (\text{Past beta}) \rightarrow$$

$$\text{Forecast beta} = a + b \times (\text{Current beta})$$

or

$$\text{Current beta} = a + b_1 \times (\text{Past beta}) + b_2 \times (\text{Firm size}) + b_3 \times (\text{Debt ratio}) \rightarrow$$

$$\text{Forecast beta} = a + b_1 \times (\text{Current beta}) + b_2 \times (\text{Firm size}) + b_3 \times (\text{Debt ratio})$$

Rosenberg and Guy (1976) found the following variables predictive:

1. Variance of earnings.
2. Variance of cash flow.
3. Growth in earnings per share.
4. Market capitalization (firm size).
5. Dividend yield.
6. Debt-to-asset ratio.

Table 8.3 Industry Betas and Adjustment Factors

Industry	Beta	Adjustment Factor
Agriculture	0.99	−0.140
Drugs and medicine	1.14	−0.099
Energy utilities	0.60	−0.237
Gold	0.36	−0.827
Construction	1.27	0.062
Air transport	1.80	0.348
Trucking	1.31	0.098
Consumer durables	1.44	0.132

Portfolio Construction and the Single-Index Model ¹

- Treynor–Black Model.
- Alpha and security analysis.
 1. Macroeconomic analysis estimates the risk premium and risk of the market index.
 2. Statistical analysis estimates beta coefficients and residual variances, $\sigma_2(e_i)$, of all securities.
 3. Establish the expected return of each security *absent any contribution from security analysis*.
 4. Various security-valuation models derive security-specific expected return forecasts.

(Adding alpha $E(R_i) = \alpha_i + \beta_i E(R_M)$)

Portfolio Construction and the Single-Index Model ₂

- Single-index model input list:
 - Risk premium on the S&P 500 portfolio.
 - Standard deviation of the S&P 500 portfolio.
 - n sets of estimates of:
 - Beta coefficients: $\beta_P = \sum_{i=1}^{n+1} w_i \beta_i$
 - Stock residual variances: $\sigma^2(e_P) = \sum_{i=1}^{n+1} w_i^2 \sigma^2(e_i)$
 - Alpha values: $\alpha_P = \sum_{i=1}^{n+1} w_i \alpha_i$

Portfolio Construction and the Single-Index Model ³

- Optimal risky portfolio in the single-index model.
- Objective is to select portfolio weights to maximize the Sharpe ratio of the portfolio.

$$E(R_P) = \alpha_P + E(R_M)\beta_P = \sum_{i=1}^{n+1} w_i \alpha_i + E(R_M) \sum_{i=1}^{n+1} w_i \beta_i$$

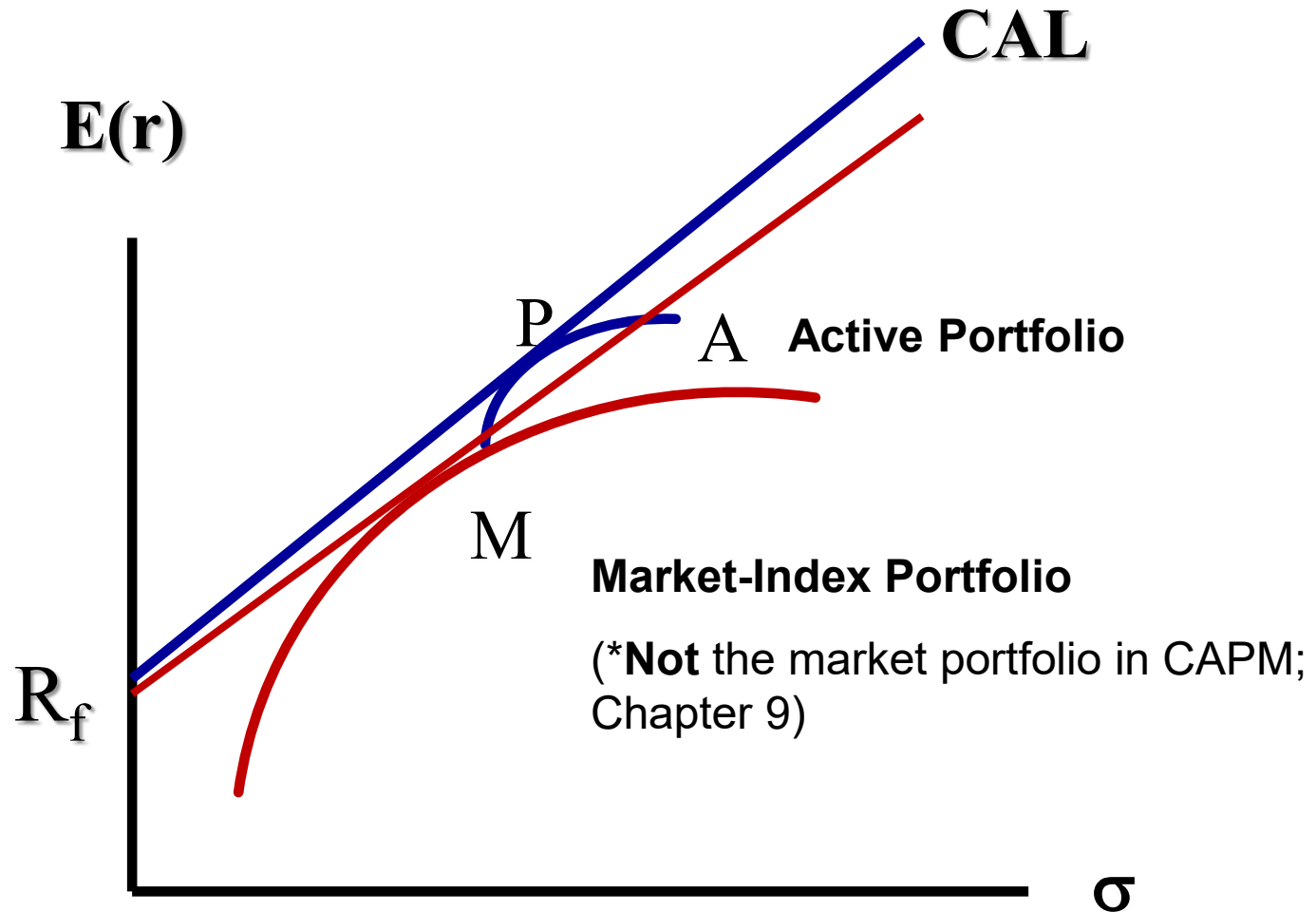
$$\sigma_P = [\beta_P^2 \sigma_M^2 + \sigma^2(e_P)]^{1/2} = \left[\sigma_M^2 \left(\sum_{i=1}^{n+1} w_i \beta_i \right)^2 + \sum_{i=1}^{n+1} w_i^2 \sigma^2(e_i) \right]^{1/2}$$

$$S_P = \frac{E(R_P)}{\sigma_P}$$

Portfolio Construction: The Process

- Optimal risky portfolio in the single-index model is a combination of two component portfolios:
 - Active portfolio, denoted by A .
 - Market-index portfolio (i.e., passive portfolio), denoted by M .
- One can use the results of Eq(7.13) to derive the optimal weight of the Active portfolio W_A (This is an optimization problem of 2 risky assets + 1 riskfree asset)

Treynor-Black Allocation



Optimal Risky Portfolio: Sharpe Ratio

- The Sharpe ratio of an optimally constructed risky portfolio will exceed that of the index portfolio (the passive strategy):

$$S_P^2 = S_M^2 + \left[\frac{\alpha_A}{\sigma(e_A)} \right]^2$$

Information Ratio of A

Optimal Risky Portfolio: Information Ratio

- The Information Ratio
 - The contribution of the active portfolio depends on the ratio of its alpha to its residual standard deviation
 - The information ratio measures the extra return we can obtain from security analysis
 - From the assumption of single index model (Refer to handout for detailed derivation)

$$S_P^2 = \frac{\left[W_A \cdot (\alpha_A + \beta_A \bar{R}_M) + (1 - W_A) \bar{R}_M \right]^2}{W_A^2 (\beta_A^2 \cdot \sigma_M^2 + \sigma^2(e_A)) + (1 - W_A)^2 \sigma_M^2 + 2W_A(1 - W_A)\beta_A\sigma_M^2} = S_M^2 + \frac{\alpha_A^2}{\sigma^2(e_A)}$$

Summary of Optimization Procedure ¹

1. Compute the initial position of each security:

$$w_i^0 = \frac{\alpha_i}{\sigma^2(e_i)}$$

2. Scale those initial positions:

$$w_i = \frac{w_i^0}{\sum_{i=1}^n w_i^0}$$

3. Compute the alpha of the active portfolio:

$$\alpha_A = \sum_{i=1}^n w_i \alpha_i$$

Summary of Optimization Procedure ²

4. Compute the residual variance of A:

$$\sigma^2(e_A) = \sum_{i=1}^n w_i^2 \sigma^2(e_i)$$

5. Compute the initial position in A:

$$w_A^0 = \frac{\alpha_A / \sigma^2(e_A)}{E(R_M) / \sigma_M^2}$$

6. Compute the beta of A:

$$\beta_A = \sum_{i=1}^n w_i \beta_i$$

Summary of Optimization Procedure ³

7. Adjust the initial position in A:

$$w_A^* = \frac{w_A^0}{1 + (1 - \beta_A) \times w_A^0}$$

Note: When $\beta_A = 1 \rightarrow w_A^* = w_A^0$

8. Optimal risky portfolio now has weights:

$$w_M^* = 1 - w_A^*$$

$$w_i^* = w_A^* \times w_i$$

Summary of Optimization Procedure ⁴

9. Calculate the risk premium of P (Optimal risky portfolio):

$$E(R_P) = (w_M^* + w_A^* \beta_A) \times E(R_M) + w_A^* \alpha_A$$

10. Compute the variance of P:

$$\sigma_P^2 = (w_M^* + w_A^* \beta_A)^2 \sigma_M^2 + [w_A^* \sigma(e_A)]^2$$

Figure 8.4 Efficient Frontier: Index Model and Full-Covariance Matrix

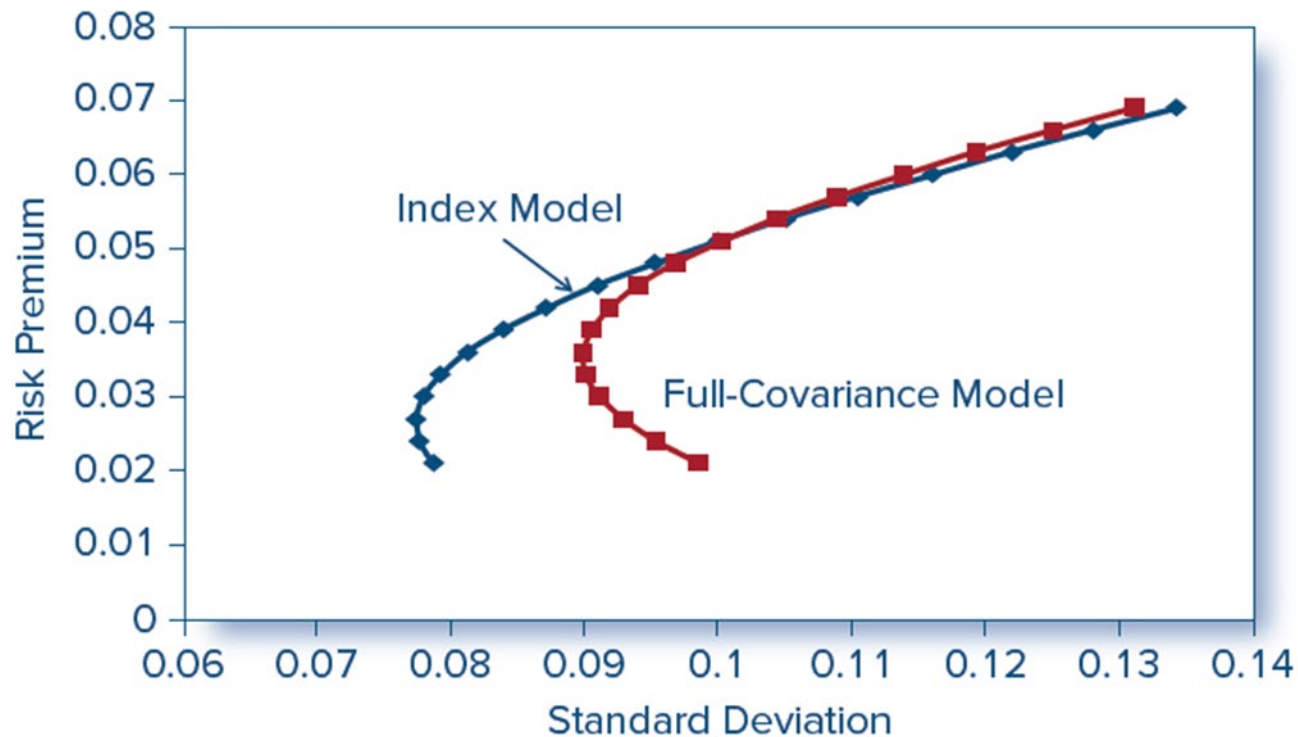


Figure 8.4 Efficient frontier constructed from the index model and the full covariance matrix

Table 8.4 Portfolios From the Index and Full-Covariance Models

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Table 8.4

Comparison of optimal risky portfolios derived from the index model and the full-covariance (Markowitz) model

	Index Model	Full-Covariance Model
A. Weights in Optimal Risky Portfolio		
Market index	0.82	0.90
WMT (Walmart)	0.13	0.17
TGT (Target)	−0.07	−0.14
VZ (Verizon)	−0.05	−0.18
T (AT&T)	0.10	0.19
F (U.S. Steel)	0.07	0.08
GM (General Motors)	0.01	−0.03
B. Portfolio Characteristics		
Risk premium	0.0605	0.0639
Standard deviation	0.1172	0.1238
Sharpe ratio	0.5165	0.5163

Is the Index Model Inferior to the Full-Covariance Model?

- Full Markowitz model is better in principle, but.
 - The full-covariance matrix invokes estimation risk of thousands of terms.
 - Cumulative errors may result in a portfolio that is actually inferior.
 - The single-index model is practical and decentralizes macro and security analysis.